

SCREENING-GRADE WATERMARK. Not an engineering design, not a field-verified parcel survey, not an observed intervention trial.

Purpose: screening-grade, funder-oriented preventive action plan (auto-filled from artifacts)

Portfolio status: counterfactual simulation - deterministic candidate suitability plus literature-based intervention effects; field verification required

Evidence labels: observed data | model output | literature assumption | counterfactual simulation

#1. Decision context

Across High Mountain Asia NOAA ISD (498 stations / 12066 station-years). Nepal-adjacent GEV: The early-period 1-in-100-year daily rainfall depth now has a fitted recurrence of 7.75 years in the late-period sample. [model output from observed NOAA ISD]

Annual-max trend: 5.229 mm/decade. [model output]

Lake-area series summary: Imja Tsho +113.3%, Thulagi +30.3%, Tsho Rolpa +26.6%. [observed/derived inventory data]

Independent check (ERA5-Land at the same station coordinates): partially replicates (16.72-yr vs ISD 7.75-yr; annmax Pearson $r=0.128$, mean bias 43.96 mm). quasi-independent: reanalysis assimilates some gauges, but it is a different measurement and modeling system from raw ISD station records [model output on reanalysis]

#2. Evidence register

- Observed data: 498 NOAA stations / 12066 station-years; UNOSAT Sentinel-1 flood extent for 2024-09-27; observed flood area 121.00 km².
- Model output: GEV return levels {'2': 29.359, '5': 41.664, '10': 49.032, '25': 57.54, '50': 63.321, '100': 68.648} with bootstrap intervals {'2': [24.821, 35.525], '5': [34.902, 50.83], '10': [42.208, 57.082], '25': [48.437, 75.369], '50': [51.13, 94.577], '100': [52.984, 120.747]}; landslide rainfall classifier, AUC out-of-sample - not a Caine-style I-D threshold with held-out AUC 0.934 on 138 events; modeled flood area 393.62 km².
- Hazard method: Copernicus GLO-30; local-min HAND proxy; CSI maximised on 27 Sep 2024 scene. This is a local-min HAND proxy calibrated on this event, not Whitebox HAND or an independently validated hydrodynamic model.
- Calibration-event metrics: CSI 0.053, POD 0.212, FAR 0.935 (event 2024-09-27; calibration-event fit, not independent validation).
- Validation table:

Event	Role	CSI	POD	FAR
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2024-09-27	in-sample calibration	0.053	0.212	0.935
2017-08-13	out-of-sample / transfer (frozen 2024 model)	0.088	0.335	0.894
2024-09-27 east half	spatial holdout (west-fit, east-test)	0.056	0.373	0.938

Baselines (same 2024 domain): JRC seasonal-water climatology CSI 0.067; area-matched elevation CSI 0.001. Permanent-water rule: JRC GSW occurrence $\geq 50\%$, 2021 v1.4 (tile 80E_30N), excluded from BOTH masks.

- Counterfactual simulation: Counterfactual exposure 77057.1 -> 54173.5 (29.7% reduction;

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simulation using observed-flood fraction x population, not observed outcomes or unique lives).

- Tail cross-check: POT/GPD return levels {'2': 230.253, '5': 282.957, '10': 324.123, '25': 380.314, '50': 424.203, '100': 469.32} [model output; different sample from the GEV headline].

- Data gap: Daily IMERG is not on an unauthenticated public URL. Use Google Earth Engine NASA/GPM_L3/IMERG_V07 or Earthdata if credentials exist. RootLedger's headline remains NOAA ISD GEV.

#3. Prioritized preventive interventions

Priority follows the optimizer's selected order. Coordinates are candidate centroids in EPSG:4326 and require site verification before procurement.

Priority	Measure ID	Measure	Coordinates (lat, lon)	Budget USD	Annual expected people-risk avoided	Risk driver	Suitability evidence	Evidence basis
1	p_c_00132_floodplain_restore	floodplain_restore	26.37000, 87.14000	32,000	2384.6	modelled_flood_exposure_and_rp100_depth	score 0.313; modeled_flood_fraction: 0.117; observed_flood_fraction: 0.037; flood_depth_rp100_m: 0.37; glof_depth_m: 0.0; landcover: cropland; +8 more fields	method: deterministic_cell_rules_v1; source_cell_id: c_00132; source_fields: ['cell_population', 'channel_proximity', 'critical_assets', 'eal_people', 'eal_usd', 'flood_depth_data_status', 'flood_depth_rp100_m', 'glof_depth_m', 'landcover', 'landcover_data_status', 'modeled_flood_fraction', 'observed_flood_fraction', 'population_data_status']; geometry_basis: hazard.geojson cell geometry; area_basis: 4.0 ha type-level screening footprint assumption; not measured from raster or cadastral data
2	p_c_00097_floodplain_restore	floodplain_restore	26.53000, 87.06000	32,000	2095.0	modelled_flood_exposure_and_rp100_depth	score 0.283; modeled_flood_fraction: 0.077; observed_flood_fraction: 0.033; flood_depth_rp100_m: 0.25; glof_depth_m: 0.0; landcover: cropland; +8 more fields	method: deterministic_cell_rules_v1; source_cell_id: c_00097; source_fields: ['cell_population', 'channel_proximity', 'critical_assets', 'eal_people', 'eal_usd', 'flood_depth_data_status', 'flood_depth_rp100_m', 'glof_depth_m', 'landcover', 'landcover_data_status', 'modeled_flood_fraction', 'observed_flood_fraction', 'population_data_status']; geometry_basis: hazard.geojson cell geometry; area_basis: 4.0 ha type-level screening footprint assumption; not measured from raster or cadastral data
3	p_c_00098_floodplain_restore	floodplain_restore	26.53000, 87.14000	32,000	2049.4	modelled_flood_exposure_and_rp100_depth	score 0.236; modeled_flood_fraction: 0.014; observed_flood_fraction: 0.006; flood_depth_rp100_m: 0.05; glof_depth_m: 0.0; landcover: cropland; +8 more fields	method: deterministic_cell_rules_v1; source_cell_id: c_00098; source_fields: ['cell_population', 'channel_proximity', 'critical_assets', 'eal_people', 'eal_usd', 'flood_depth_data_status', 'flood_depth_rp100_m', 'glof_depth_m', 'landcover', 'landcover_data_status', 'modeled_flood_fraction', 'observed_flood_fraction', 'population_data_status']; geometry_basis: hazard.geojson cell geometry; area_basis: 4.0 ha type-level screening footprint assumption; not measured from raster or cadastral data
4	p_c_00115_floodplain_restore	floodplain_restore	26.45000, 87.14000	32,000	1956.5	modelled_flood_exposure_and_rp100_depth	score 0.293; modeled_flood_fraction: 0.091; observed_flood_fraction: 0.012; flood_depth_rp100_m: 0.29; glof_depth_m: 0.0; landcover: cropland; +8 more fields	method: deterministic_cell_rules_v1; source_cell_id: c_00115; source_fields: ['cell_population', 'channel_proximity', 'critical_assets', 'eal_people', 'eal_usd', 'flood_depth_data_status', 'flood_depth_rp100_m', 'glof_depth_m', 'landcover', 'landcover_data_status', 'modeled_flood_fraction', 'observed_flood_fraction', 'population_data_status']; geometry_basis: hazard.geojson cell geometry; area_basis: 4.0 ha

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type-level screening footprint assumption; not measured from raster or cadastral data |
| 5 | p_c_00100_floodplain_restore | floodplain_restore | 26.53000, 87.30000 | 32,000 |
1756.3 | modelled_flood_exposure_and_rp100_depth | score 0.226; modeled_flood_fraction:
0.001; observed_flood_fraction: 0.0; flood_depth_rp100_m: 0.05; glof_depth_m: 0.0;
landcover: cropland; +8 more fields | method: deterministic_cell_rules_v1; source_cell_id:
c_00100; source_fields: ['cell_population', 'channel_proximity', 'critical_assets',
'eal_people', 'eal_usd', 'flood_depth_data_status', 'flood_depth_rp100_m', 'glof_depth_m',
'landcover', 'landcover_data_status', 'modeled_flood_fraction', 'observed_flood_fraction',
'population_data_status']; geometry_basis: hazard.geojson cell geometry; area_basis: 4.0 ha
type-level screening footprint assumption; not measured from raster or cadastral data |
| 6 | p_c_00133_floodplain_restore | floodplain_restore | 26.37000, 87.22000 | 32,000 |
1539.1 | modelled_flood_exposure_and_rp100_depth | score 0.299; modeled_flood_fraction:
0.099; observed_flood_fraction: 0.042; flood_depth_rp100_m: 0.32; glof_depth_m: 0.0;
landcover: cropland; +8 more fields | method: deterministic_cell_rules_v1; source_cell_id:
c_00133; source_fields: ['cell_population', 'channel_proximity', 'critical_assets',
'eal_people', 'eal_usd', 'flood_depth_data_status', 'flood_depth_rp100_m', 'glof_depth_m',
'landcover', 'landcover_data_status', 'modeled_flood_fraction', 'observed_flood_fraction',
'population_data_status']; geometry_basis: hazard.geojson cell geometry; area_basis: 4.0 ha
type-level screening footprint assumption; not measured from raster or cadastral data |
| 7 | p_c_00114_floodplain_restore | floodplain_restore | 26.45000, 87.06000 | 32,000 |
1359.8 | modelled_flood_exposure_and_rp100_depth | score 0.297; modeled_flood_fraction:
0.096; observed_flood_fraction: 0.028; flood_depth_rp100_m: 0.31; glof_depth_m: 0.0;
landcover: cropland; +8 more fields | method: deterministic_cell_rules_v1; source_cell_id:
c_00114; source_fields: ['cell_population', 'channel_proximity', 'critical_assets',
'eal_people', 'eal_usd', 'flood_depth_data_status', 'flood_depth_rp100_m', 'glof_depth_m',
'landcover', 'landcover_data_status', 'modeled_flood_fraction', 'observed_flood_fraction',
'population_data_status']; geometry_basis: hazard.geojson cell geometry; area_basis: 4.0 ha
type-level screening footprint assumption; not measured from raster or cadastral data |
| 8 | p_c_00080_floodplain_restore | floodplain_restore | 26.61000, 87.06000 | 32,000 |
1245.5 | modelled_flood_exposure_and_rp100_depth | score 0.227; modeled_flood_fraction:
0.003; observed_flood_fraction: 0.003; flood_depth_rp100_m: 0.05; glof_depth_m: 0.0;
landcover: cropland; +8 more fields | method: deterministic_cell_rules_v1; source_cell_id:
c_00080; source_fields: ['cell_population', 'channel_proximity', 'critical_assets',
'eal_people', 'eal_usd', 'flood_depth_data_status', 'flood_depth_rp100_m', 'glof_depth_m',
'landcover', 'landcover_data_status', 'modeled_flood_fraction', 'observed_flood_fraction',
'population_data_status']; geometry_basis: hazard.geojson cell geometry; area_basis: 4.0 ha
type-level screening footprint assumption; not measured from raster or cadastral data |
| 9 | p_c_00081_floodplain_restore | floodplain_restore | 26.61000, 87.14000 | 32,000 |
1177.2 | modelled_flood_exposure_and_rp100_depth | score 0.229; modeled_flood_fraction:
0.005; observed_flood_fraction: 0.0; flood_depth_rp100_m: 0.05; glof_depth_m: 0.0;
landcover: cropland; +8 more fields | method: deterministic_cell_rules_v1; source_cell_id:
c_00081; source_fields: ['cell_population', 'channel_proximity', 'critical_assets',
'eal_people', 'eal_usd', 'flood_depth_data_status', 'flood_depth_rp100_m', 'glof_depth_m',
'landcover', 'landcover_data_status', 'modeled_flood_fraction', 'observed_flood_fraction',
'population_data_status']; geometry_basis: hazard.geojson cell geometry; area_basis: 4.0 ha
type-level screening footprint assumption; not measured from raster or cadastral data |
| 10 | p_c_00116_floodplain_restore | floodplain_restore | 26.45000, 87.22000 | 32,000 |
1104.2 | modelled_flood_exposure_and_rp100_depth | score 0.231; modeled_flood_fraction:
0.008; observed_flood_fraction: 0.003; flood_depth_rp100_m: 0.05; glof_depth_m: 0.0;
landcover: cropland; +8 more fields | method: deterministic_cell_rules_v1; source_cell_id:
c_00116; source_fields: ['cell_population', 'channel_proximity', 'critical_assets',
'eal_people', 'eal_usd', 'flood_depth_data_status', 'flood_depth_rp100_m', 'glof_depth_m',

'landcover', 'landcover_data_status', 'modeled_flood_fraction', 'observed_flood_fraction', 'population_data_status']; geometry_basis: hazard.geojson cell geometry; area_basis: 4.0 ha type-level screening footprint assumption; not measured from raster or cadastral data | 11 | p_c_00069_floodplain_restore | floodplain_restore | 26.61000, 86.18000 | 32,000 | 858.7 | modelled_flood_exposure_and_rp100_depth | score 0.248; modeled_flood_fraction: 0.031; observed_flood_fraction: 0.013; flood_depth_rp100_m: 0.1; glof_depth_m: 0.0; landcover: cropland; +8 more fields | method: deterministic_cell_rules_v1; source_cell_id: c_00069; source_fields: ['cell_population', 'channel_proximity', 'critical_assets', 'eal_people', 'eal_usd', 'flood_depth_data_status', 'flood_depth_rp100_m', 'glof_depth_m', 'landcover', 'landcover_data_status', 'modeled_flood_fraction', 'observed_flood_fraction', 'population_data_status']; geometry_basis: hazard.geojson cell geometry; area_basis: 4.0 ha type-level screening footprint assumption; not measured from raster or cadastral data | 12 | p_c_00096_floodplain_restore | floodplain_restore | 26.53000, 86.98000 | 32,000 | 720.7 | modelled_flood_exposure_and_rp100_depth | score 0.238; modeled_flood_fraction: 0.007; observed_flood_fraction: 0.017; flood_depth_rp100_m: 0.05; glof_depth_m: 0.0; landcover: cropland; +8 more fields | method: deterministic_cell_rules_v1; source_cell_id: c_00096; source_fields: ['cell_population', 'channel_proximity', 'critical_assets', 'eal_people', 'eal_usd', 'flood_depth_data_status', 'flood_depth_rp100_m', 'glof_depth_m', 'landcover', 'landcover_data_status', 'modeled_flood_fraction', 'observed_flood_fraction', 'population_data_status']; geometry_basis: hazard.geojson cell geometry; area_basis: 4.0 ha type-level screening footprint assumption; not measured from raster or cadastral data |

Measure	Sites	Modeled spend USD	Annual expected people-risk avoided
floodplain restore	62	1,984,000	26306.1

- ## #5. Assumptions and evidence basis

- vetiver_slope - cost 5,000 USD/ha; effect fraction 0.35; carbon 12.0 tCO₂/ha/10yr; income 600 USD/ha/yr. Source: Vetiver hedgerows cut slope erosion >90% (Truong et al.); cost/income from ICIMOD bioengineering field costings. [literature assumption]
- bamboo_slope - cost 4,000 USD/ha; effect fraction 0.30; carbon 45.0 tCO₂/ha/10yr; income 850 USD/ha/yr. Source: Bamboo root reinforcement for shallow landslides; bamboo is an established Nepali rural cash crop (INBAR). [literature assumption]
- floodplain_restore - cost 8,000 USD/ha; effect fraction 0.30; carbon 35.0 tCO₂/ha/10yr; income 300 USD/ha/yr. Source: NbS flood-depth reduction ~15-40% (UNEP); floodplain reconnection costs from World Bank NbS catalogue. [literature assumption]
- wetland_restore - cost 12,000 USD/ha; effect fraction 0.35; carbon 60.0 tCO₂/ha/10yr; income 400 USD/ha/yr. Source: Wetland attenuation of peak flows; high soil-carbon uptake (Ramsar / IPCC wetlands supplement). [literature assumption]
- afforestation - cost 3,000 USD/ha; effect fraction 0.20; carbon 80.0 tCO₂/ha/10yr; income

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250 USD/ha/yr. Source: Catchment afforestation for slope stability and infiltration; sequestration from IPCC AFOLU defaults. [literature assumption]

- riverbank_bio - cost 6,000 USD/ha; effect fraction 0.25; carbon 20.0 tCO₂/ha/10yr; income 500 USD/ha/yr. Source: Bioengineered riverbank protection (live crib walls, brush layering) per ICIMOD/DSCWM manuals. [literature assumption]

The optimizer applies marginal greedy triple-return per dollar, per-cell EAL capping, and Monte-Carlo climate multipliers derived from the GEV 100-year confidence interval. [model design assumption]

Equity weighting (verbatim from hazard/proof.py _grid_features): rural = landcover in cropland/grass/shrub AND no critical_assets; dense = population at or above the 70th percentile of cell pops; low_income_score = 1.0 if (rural and dense) else (0.55 if rural else 0.15); equity_weight = 1.0 + 0.5 * low_income_score (range 1.075-1.5x). eal_people is multiplied by equity_weight, so low-income rural exposure ranks higher in expected-loss. [model design assumption]

No causal percentage is assigned to government, communities, or households. Implementation roles require local governance and community confirmation.

#6. Monitoring and verification

- Rainfall signal: append observed NOAA ISD records annually; re-fit GEV and publish sample size, confidence interval, and any changed recurrence estimate.
- Flood extent: after qualifying events, derive observed Sentinel-1 extent and report POD, FAR, and CSI separately from the calibration event.
- Intervention delivery: verify each parcel centroid, boundary, measure type, area, and installation date through field sign-off plus dated optical imagery.
- Carbon: report planted/restored area and survival first; convert to tCO₂ only with the stated factor and uncertainty. Ground-sample biomass before claiming credits.
- Risk performance: compare post-event observed exposure with a pre-registered baseline; do not present the counterfactual simulation as an observed outcome.
- Livelihoods: measure participant payments or income directly before making beneficiary, household, job, or wage claims.

#7. Limitations

- The flood surface is a screening-grade local-min HAND proxy calibrated on the 27 Sep 2024 event. Its event-fit CSI is low and is not independent validation.
- Hazard depths, EAL, avoided people-risk, carbon, income, and the optimized ranking are model outputs; field feasibility, tenure, ecological safeguards, and engineering design remain unverified.
- Candidate centroids are screening locations, not surveyed parcel boundaries.
- Literature effect sizes and unit costs may not transfer to local implementation conditions.
- The counterfactual is a simulation, not a prediction or measured causal effect.
- IMERG was not fused, and the HMA-pooled late GEV did not identify the Nepal-adjacent headline shift.